

What a Filing Reveals: A Companion Model for the Electoral Costs of Judicialization

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Abstract

Adversarial electoral litigation widens the top-two margin in Brazilian mayoral races. This note asks what mechanism can produce that, and finds that the natural candidate cannot. In the transfer class of campaign-attack models, an attack moves support from its target to its sender; under the attack direction those models derive and document, the map is a contraction on the top-two margin, so it cannot generate the result. The alternative developed here turns on what a filing reveals. A filing is a noisy signal about a candidate's type and a precise signal about a candidate's rank, because the filer is optimizing and targets the leading opponent. Litigation therefore informs voters about position more than about quality. Coordination on the revealed leader widens the margin; the collapse of pivot probability sends some of the runner-up's support off the valid ballot, which under compulsory voting means a blank or null vote. Whether that support joins the leader or leaves depends on how settled the leader's reputation is, which is what separates open from contested seats.

1 Environment

One mayoral race. Candidates $j = 1, \dots, J$ are indexed by pre-shock support $s_1^\circ \geq s_2^\circ \geq \dots \geq s_J^\circ$, and there is a continuum of voters of mass one. Voter i 's payoff from candidate j is

$$u_{ij} = \hat{\theta}_j - \frac{\rho}{2} \hat{V}_i(\theta_j) + \varepsilon_{ij},$$

where θ_j is a common-value valence nobody observes and ε_{ij} is a fixed idiosyncratic taste the voter knows. Beliefs about valence start at $\theta_j \sim N(m_j, 1/h_j)$: m_j is the candidate's reputation and h_j measures how settled the record is. An incumbent who has governed for four years has a high h ; a first-time candidate has a low one. Let $\bar{b}$ denote the payoff from a blank or null vote, which is what refusing to endorse anyone is worth once the ballot carries no instrumental value.

Treatment is λ , the intensity of adversarial filings in the race, and it is the object the shift-share instrument shifts. Two institutional facts fix the timing and the margin of adjustment. Candidate registration closes on 15 August while adversarial filings run through October, so the candidate set is predetermined with respect to λ . And voting is compulsory (CF/88 art. 14 §1), so a voter who wants out of the contest adjusts the mark rather than the attendance: the exit option is a blank or null vote.

2 The transfer channel cannot produce the result

2.1 The map

Nakaguma and Souza (2025) model a campaign attack as a transfer. An attack from k on j

moves a fraction ϕ of j 's support away from j ; in their Appendix G.2 demobilization extension a fraction α of that mass leaves for the outside option instead of reaching k , and the attacker loses a fraction γ of its own support to backlash.

Their equilibrium has every candidate target the highest-ranked opponent, so in a three-candidate race the profile is $1 \rightarrow 2$, $2 \rightarrow 1$ and $3 \rightarrow 1$: mutual between the top two, with the third attacking the leader. Their data agrees. Measuring attacks as electoral lawsuits, the observed frequencies in three-candidate races are $2 \rightarrow 1$ at 3.0pp, $1 \rightarrow 2$ at 1.9pp and $3 \rightarrow 1$ at 0.8pp. The second-place candidate is the most aggressive and the front-runner is the most attacked.

2.2 The contraction

Proposition 1 (Transfer cannot consolidate). *Under mutual $1 \leftrightarrow 2$ attacks, post-attack support is*

$$s_1 = s_1^\circ(1 - \phi - \gamma) + \phi(1 - \alpha)s_2^\circ, \quad s_2 = s_2^\circ(1 - \phi - \gamma) + \phi(1 - \alpha)s_1^\circ,$$

so the top-two margin satisfies

$$s_1 - s_2 = (s_1^\circ - s_2^\circ)\kappa, \quad \kappa \equiv 1 - \gamma - \phi(2 - \alpha).$$

For every $\phi \in (0, 1)$, $\alpha \in [0, 1]$ and $\gamma \in [0, 1)$ we have $\kappa < 1$. Adding $3 \rightarrow 1$ subtracts a further ϕs_1° from the margin. Hence the transfer map cannot widen the top-two margin while preserving the identity of the leader.

Proof. Subtracting,

$$s_1 - s_2 = (s_1^\circ - s_2^\circ)(1 - \phi - \gamma) - \phi(1 - \alpha)(s_1^\circ - s_2^\circ) = (s_1^\circ - s_2^\circ)[1 - \gamma - \phi(2 - \alpha)].$$

Since $\alpha \leq 1$ we have $2 - \alpha \geq 1$, so $\phi(2 - \alpha) \geq \phi > 0$, and with $\gamma \geq 0$ this gives $\kappa < 1$. If $\kappa \in (0, 1)$ the margin strictly narrows. If $\kappa \leq 0$ the sign of $s_1 - s_2$ is non-positive, so candidate 2 has caught or passed candidate 1 and the leader is no longer the pre-attack leader. When $3 \rightarrow 1$ is added, candidate 1 loses a further ϕs_1° and candidate 2 loses nothing, so $s_1 - s_2 = (s_1^\circ - s_2^\circ)\kappa - \phi s_1^\circ$, which is smaller still. $\square$

Remark. Widening the margin under the transfer map requires a one-sided downward attack, $1 \rightarrow 2$ with no reciprocation. That is excluded by the equilibrium, in which candidate 2's best target is candidate 1, and it is not what the data shows. The per-pair attack frequencies are also 1–3pp, an order of magnitude too small to move municipality-level vote shares by three points.

Remark (Object of comparison). Nakaguma and Souza (2025) work with a win probability through a logit contest function, whereas the estimates here are vote-share margins. The two are monotone in $s_1 - s_2$, so Proposition 1 transfers, but the mapping is an assumption and not an identity.

Remark (What Proposition 1 does not do). The top-two margin is an order statistic, so any mean-preserving increase in the dispersion of vote shares widens it mechanically. Proposition 1 rules out the transfer channel under the documented attack direction; it does not rule out a generic dispersion shock. What rules that out is the ballot response and the contrast between seat types in Section 5: a dispersion shock predicts neither, and predicts no directional flow from the runner-up to the leader.

3 What a filing emits

A filing carries two signals, and their precisions differ by an order of magnitude.

The type signal. A filing is an allegation about θ_j . It is informative with probability π and pure noise with probability $1 - \pi$, and the voter cannot tell which. Writing τ_I for the precision of an informative filing, the effective precision is $\bar{\tau} = \pi\tau_I$, and the posterior mean is

$$\hat{\theta}_j = \frac{h_j m_j + \bar{\tau} s_j}{h_j + \bar{\tau}}.$$

Genuine information about a candidate and a weapon that happens to persuade enter through the same $\bar{\tau}$. They are therefore not separable for the voter, and not separable for the econometrician either. The non-separability that the empirical design has to live with is a property of the environment, not a shortcoming of the instrument (Feddersen and Pesendorfer, 1996).

The status signal. The filer is optimizing and targets the highest-ranked opponent, which is the premise Proposition 1 takes from Nakaguma and Souza (2025). So the identity of the target is informative about who is ahead, and its precision is high, because the filer knows the race. A court case also draws local attention, and revealing which candidates are first and second is known to move votes (Anagol and Fujiwara, 2016).

Litigation is thus precise about position and imprecise about quality.

4 Consolidation

Proposition 2 (Consolidation). *As λ rises, rank moves toward common knowledge while valence beliefs barely update. In a coordination equilibrium over the two leading candidates (Myerson and Weber, 1993; Fey, 1997; Cox, 1997), support flows from the runner-up to the leader. The identity of the leader is unchanged and the size of the field is unchanged.*

The candidate-supply null is structural rather than incidental: the status signal arrives after the registration deadline, so it cannot act on the candidate set.

Proposition 3 (Concentration indices are redundant). *If the shift is a one-parameter reallocation between the top two, then every summary concentration index – the effective number of candidates, the Herfindahl index, the combined top-two share – is a monotone function of that same parameter, and none of them carries a restriction independent of the margin.*

This is why concentration cannot identify which mechanism is at work. The point is not that the concentration estimates are imprecise; it is that even a precise one would be the margin restated. Pushing the estimated top-two reallocation through each municipality’s own 2024 share vector implies $\Delta\text{HHI} = +0.018$ and $\Delta\text{ENP} = -0.053$, and the separately estimated index regressions give $+0.023$ (0.013) and -0.044 (0.039). Both implied values sit inside the estimated intervals, so the index regressions add no restriction to the margin result (docs/model/check_proposition2.py).

5 Exit and the seat split

Proposition 4 (Exit). *A supporter of the runner-up who learns the race is not close faces a collapsed pivot probability, so the choice becomes expressive: mark the leader if $\hat{\theta}_1 + \varepsilon_{i1} \geq \bar{b}$, and otherwise mark blank. Under compulsory voting the exit option is a blank or null vote rather than abstention, so turnout is unmoved (Ghirardato and Katz, 2006).*

Proposition 5 (The seat split). *The runner-up loses a similar mass whichever kind of seat is contested. What differs is where that mass goes: to the leader where the revealed leader is a fresh candidate, and off the valid ballot where the revealed leader is an incumbent.*

Two primitives differ across seat types, and both push the same way.

First, the prior precision h_1 . In a contested seat the leader is the incumbent, h_1 is high, $\hat{\theta}_1 \approx m_1$, and the type signal moves nothing. In an open seat the leader is a fresh candidate with a diffuse reputation and a genuinely unresolved rank, so the status signal has more to resolve and coordination is stronger.

Second, selection on ε_{i1} . Supporting the challenger against a sitting mayor is itself evidence of a settled negative view of that mayor, so the distribution of ε_{i1} among the runner-up’s supporters is first-order dominated in contested seats. A larger mass sits below $\bar{b}$ and exits. An open-seat leader has not been in office long enough to have accumulated committed opponents.

6 What the model does not do

The model produces signs and orderings. It is not calibrated and not structurally estimated. It does not separate a genuine allegation from a persuasive weapon, and Section 3 is the reason that is a property of the environment rather than a gap to be closed. It does not endogenize the filer’s decision: the attack direction is taken from Nakaguma and Souza (2025) as a maintained assumption, because endogenizing it would require exactly the selection the instrument exists to sidestep. It says nothing about judges, and nothing about welfare. Runoff municipalities, where a second round changes the coordination problem, are outside it.

Two conditions bound the mechanism. Coordination needs a small number of focal contenders, so a genuinely three-way race weakens it. And the reasoning in Section 4 applies to a single-winner plurality contest; an open-list proportional council race offers no focal rank for the status signal to reveal, which is the model’s clearest out-of-sample implication.

References

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